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(B.9) $c_1 \underline{x}^{(1)} + c_2 \underline{x}^{(2)} + \cdots c_k \underline{x}^{(k)} = \underline{0}$

$$c_1 \begin{pmatrix} x_1^{(1)} \\ x_2^{(1)} \\ \vdots \\ x_n^{(1)} \end{pmatrix} + c_2 \begin{pmatrix} x_1^{(2)} \\ x_2^{(2)} \\ \vdots \\ x_n^{(2)} \end{pmatrix} + \cdots + c_k \begin{pmatrix} x_1^{(k)} \\ x_2^{(k)} \\ \vdots \\ x_n^{(k)} \end{pmatrix} = \underline{0}$$

$$= \underbrace{\begin{pmatrix} x_1^{(1)} & x_1^{(2)} & \cdots & x_1^{(k)} \\ \vdots & & & \\ x_n^{(1)} & - & - & x_n^{(k)} \end{pmatrix}}_{\underline{\underline{X}}} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_k \end{pmatrix} = \underline{0}$$

(B.10) $\underline{\underline{X}}\, \underline{c} = \underline{0}$

$$\underline{\underline{X}} = \begin{pmatrix} 1 & 2 & -4 \\ 2 & 1 & 1 \\ -1 & 3 & -11 \end{pmatrix} \qquad |\underline{\underline{X}}| = 0$$

$\underline{x}^{(1)}$ $\underline{x}^{(2)}$ $\underline{x}^{(3)}$

Solve $\underline{\underline{X}}\, \underline{c} = 0$ to find $\underline{c}$

$$\begin{pmatrix} 1 & 2 & -4 \\ 2 & 1 & 1 \\ -1 & 3 & -11 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \underline{0}$$

either $c_1 + 2c_2 - 4c_3 = 0$
— $2c_1 + c_2 + c_3 = 0$
$-c_1 + 3c_2 - 11c_3 = 0$

or use row operations

$\therefore\ 2\underline{x}^{(1)} - 3\underline{x}^{(2)} - \underline{x}^{(3)} = \underline{0}$

Recall $\underline{\underline{A}}\, \underline{x} = \lambda \underline{x} \quad \Rightarrow |A - \lambda \underline{\underline{I}}| = 0$

$n \times n$; $\lambda_1, \lambda_2, \ldots \lambda_n$

$n \times n$: $\underline{\underline{P}} = \begin{pmatrix} \underline{x}^{(1)} & \underline{x}^{(2)} & \cdots & \underline{x}^{(n)} \end{pmatrix}$

eigenvectors

$$\underline{\underline{A}}\,\underline{\underline{P}} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & & & \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} \underline{x}^{(1)} & \underline{x}^{(2)} & \cdots & \underline{x}^{(n)} \end{pmatrix} = \begin{pmatrix} A\underline{x}^{(1)} & A\underline{x}^{(2)} & \cdots & A\underline{x}^{(n)} \end{pmatrix}$$

$$= \begin{pmatrix} \lambda_1 \underline{x}^{(1)} & \lambda_2 \underline{x}^{(2)} & \cdots \end{pmatrix}$$

$\underline{\underline{A}}\,\underline{\underline{P}} = \underline{\underline{P}}\,\underline{\underline{D}}$ ← diagonal matrix of eigenvalues

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Activity 8.8 $A = \begin{pmatrix} 5 & -1 \\ 3 & 1 \end{pmatrix}$ find $\underbrace{\lambda_1, \lambda_2}_{D}$ $\underbrace{\underline{x}^{(1)}, \underline{x}^{(2)}}_{P}$

a_{22}

$a_{22}a_{11}x_1 + a_{12}x_2 = b_1 a_{22}$

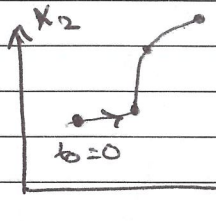
a_{12}

$a_{12}a_{21}x_1 + a_{22}x_2 = b_2 a_{12}$

$a_{22}a_{11}x_1 - a_{12}a_{21}x_1 + 0 = b_1 a_{22} - b_2 a_{12}$

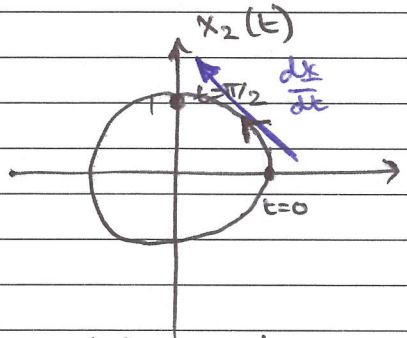
$\therefore x_1 = \dfrac{b_1 a_{22} - b_2 a_{12}}{a_{22}a_{11} - a_{12}a_{21} = \Delta}$

$\underline{A}\underline{x} = \underline{b}$ $\qquad \underbrace{\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}}_{A} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$



parametric curve, parameter t

Example 4.1 $\underline{x}(t) = \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}$ $\qquad \dfrac{d\underline{x}}{dt} = \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix}$



@ $t=0$ $\underline{x}(0) = \begin{pmatrix} \cos 0 \\ \sin 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

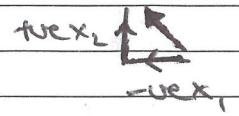
@ $t=\pi/2$ $\underline{x}(\pi/2) = \begin{pmatrix} \cos \pi/2 \\ \sin \pi/2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

traversed anticlockwise

$x_1(t) = \cos t$

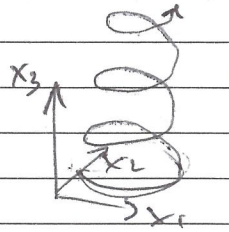
$x_2(t) = \sin t$

$x_1^2(t) + x_2^2(t) = \cos^2 t + \sin^2 t = 1$ circle radius 1



$\left.\dfrac{d\underline{x}}{dt}\right|_{\pi/4} = \begin{pmatrix} -\sin \pi/4 \\ \cos \pi/4 \end{pmatrix} = \begin{pmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$

$\underline{x} = \begin{pmatrix} \cos t \\ \sin t \\ t \end{pmatrix}$



helix

Example 4.6 $\quad \frac{dx_1}{dt} = x_1(t) + x_2(t)$

$$\frac{dx_2}{dt} = 4x_1(t) + x_2(t)$$

$$\frac{d\underline{x}}{dt} = P(t)\,\underline{x}$$

let $\underline{x} = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}$ then $\frac{d\underline{x}}{dt} = \frac{d}{dt}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}}_{P}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

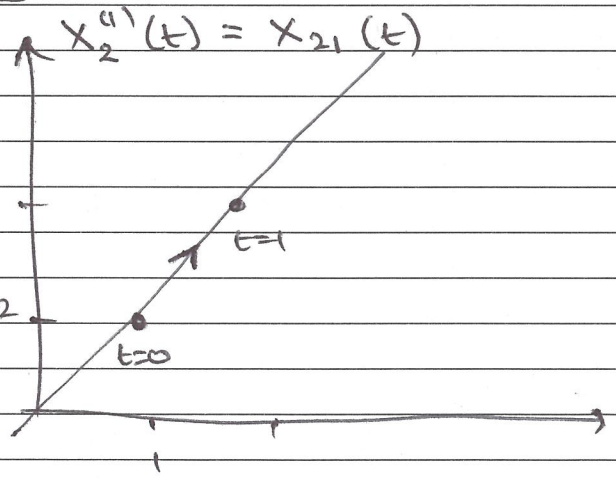
(i) Take $\underline{x}^{(1)}(t) = \begin{pmatrix} e^{3t} \\ 2e^{3t} \end{pmatrix}$ then $\frac{d\underline{x}^{(1)}}{dt} = \begin{pmatrix} 3e^{3t} \\ 6e^{3t} \end{pmatrix}$

whilst $P\underline{x}^{(1)} = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}\begin{pmatrix} e^{3t} \\ 2e^{3t} \end{pmatrix}$

$$= \begin{pmatrix} e^{3t} + 2e^{3t} \\ 4e^{3t} + 2e^{3t} \end{pmatrix} = e^{3t}\begin{pmatrix} 3 \\ 6 \end{pmatrix}$$

$\therefore$ LHS = RHS and $\underline{x}^{(1)}$ is a solution ✓

(ii) Check $\underline{x}^{(2)}(t)$ is a sol^{n}.



$$\underline{x}^{(1)} = \begin{pmatrix} e^{3t} \\ 2e^{3t} \end{pmatrix}$$

@ $t=0$ $\quad \underline{x}^{(1)} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

@ $t=1$ $\quad \underline{x}^{(1)} = \begin{pmatrix} e^3 \\ 2e^3 \end{pmatrix}$

$x_1^{(1)}(t) = e^{3t}$

$x_2^{(1)}(t) = 2e^{3t} = 2x_1^{(1)}(t)$ straight line with slope 2.